

BNY PLACEMENT DRIVE

- JOSHIKA SHREE V

BNY (The Bank of New York Corporation) is a leading American multinational financial services company headquartered in New York City. Recognized as one of the world's largest custodian banks, it specializes in investment management, asset servicing, and wealth management, empowering institutions and individuals to achieve success in global markets.

With a rich history spanning over 240 years, BNY Mellon has built a reputation for trust, innovation, and client-centric solutions. The company continues to shape the future of finance by combining deep expertise with advanced technology, helping businesses and individuals thrive in a rapidly evolving financial world.

Overall process:

- *Online Test*
 - *Technical Interview(@CUIC)*
 - *Technical + HR Interview(@CUIC)*
-

1.ONLINE TEST:

The first stage was a pure coding round conducted on the HackerRank platform, with a total duration of 3 hours. The test consisted of 4 programming questions

- *1 Easy*
- *2 Moderate*
- *1 Hard*

The test allowed candidates to code in any programming language of their choice.

1. *Developer Team Formation: Each developer has skill levels from 1 to n, and limits on how many teammates with lower or higher skills they can work with. A developer is “content” if these limits are satisfied. The task was to form the largest possible team where everyone is content.*

Example: $n=3$, $\text{lowerSkill}=[0,1,2]$,

$\text{higherSkill}=[2,1,0] \rightarrow \text{Team } \{1,2,3\}$

works because all developers' limits are satisfied $\rightarrow$ largest team size = 3.

2. *Largest Subset of Events: In a scheduling system, each event has a start and end time. A subset of events is called high-priority if every event overlaps or touches every other event in the subset. The task was to find the maximum-size high-priority subset.*

Example: $\text{start}=[1,2,3,4]$

$\text{finish}=[2,3,5,6] \rightarrow \text{Subset } \{2,3,4\}$

overlaps fully $\rightarrow$ largest subset size = 3.

3. *Minimum Node Types in a Graph: Given an undirected graph, the goal was to assign a type to each node so that:*

- Connected nodes have different types*
 - Nodes sharing a neighbor also have different types*
- The objective was to minimize the number of types used.*

Example: $n=4$

$\text{edges}=[(1,2),(2,3),(2,4),(3,4)] \rightarrow \text{Minimum types required} = 3.$

4. *Server State Flipping: A binary string represented server states ('1' = active, '0' = inactive), and a list of operations defined ranges to flip. The task was to find how many distinct active server counts can be obtained by applying any subset of operations.*

Example: $\text{servers}="101"$,

$\text{operations}=[(0,1),(1,2)] \rightarrow \text{Possible active counts} = \{1,2\}.$

I successfully completed two problems, nearly completed one (14/15 test cases), and partially completed another (3/15 test cases)

2. TECHNICAL INTERVIEW :(45-50 minutes)

The round started with a quick self-introduction. Immediately, the focus shifted to my projects. The interviewer asked several in-depth questions about them.

Example: "Can you explain the logic behind authentication and authorization?"

Since the company was finance-based, the interviewer gave special importance to DBMS concepts (like locks and related concepts).

- *To explain the collections used in my MERN stack projects along with their attributes.*
- *To solve around 5 SQL queries based on my project design.*

Tip : Be very clear about your projects — not just features, but also data flow, database schema, and logic behind implementation.

Apart from projects, I faced theory-based questions from different subjects:

- *OOPs: Difference between Abstraction and Interface (with real-life examples).*
- *SDLC:*
 - *"Where would you integrate AI in your project?"*
 - *"How does AI fit into the five phases of SDLC?"*
- *OS basics and general software engineering fundamentals.*

Tip: Don't just memorize definitions — learn to apply concepts to real-world cases. That's what interviewers look for.

DSA Coding Problems

Finally, I was asked to write codes for:

- *BFS Traversal*
- *Dijkstra's Shortest Path Algorithm*
- *Linked List Node Deletion*

Tip: Always stay strong in DSA that's gonna help u a lot!

QUICK TAKEAWAYS:

- *Proficient and consistently trained in DSA*
- *Thoroughly confident in projects and backend design*
- *Solid understanding of DBMS, OOPs, OS, and core fundamentals*
- *Skilled at handling complex queries under pressure*
- *Communicate solutions with clarity and confidence*

3. TECHNICAL and HR INTERVIEW (30-40 Minutes):

Within just 10 minutes, I was called for the next round. This one started chill, again with the usual self-introduction and project discussion.

This round had some comparatively tougher queries than the previous one. I was asked to explain the difference between RANK() and DENSE_RANK(), followed by a few more advanced SQL queries.

One of the questions was how to convert a row-oriented table into a column-oriented format, for which I explained the approach using window functions.

Next, I was given a coding problem: an expense splitter/tracker. The problem was:

- *A contributes 1200, B contributes 800, C contributes 1000 for a trip.*
- *After calculating each person's fair share, I was asked to determine who should give money to whom and how much.*
- *For this example: A had paid 200 extra, so the system should return 200 to A from B, while C's contribution remained the same.*

After solving this, a few reasoning and math questions were asked. For instance:

- *If you start with 100 Rs, gain 10% profit at the end of the day, and 10% loss the next day, what is the final loss?*
- *Then comes the HR section including behavioral questions, like why the interviewer shouldn't hire me, along with the usual HR questions. I also took the chance to highlight my college volunteering and leadership experiences, adding a small, confident highlight of my achievements.*

QUICK TAKEAWAYS:

Mental Math & Reasoning: Quick thinking matters.

Be creative: Avoid standard, safe responses; think outside the box.

Show, don't tell: Instead of saying "I can do this," demonstrate your skills indirectly through your examples, experiences, and the way you speak.

Interviews aren't just about hardwork and effort — they're about timing and fit.

What's meant for you will definitely find you. Until then keep learning and stay confident

You're already halfway to victory — now keep going. You've got this..

Your success story is waiting to be told!!!!

ALL THE VERY BEST :)

PING ME ANYTIME- 8838381339